

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2     3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(cowplot)
```

```
##  
## Attaching package: 'cowplot'  
##  
## The following object is masked from 'package:lubridate':  
##  
## stamp
```

```
library(lubridate)
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
processed_data = here("Data/Processed_KEY")  
PeterPaul.chem.nutrients <- read.csv(  
  here(processed_data, "NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),  
  stringsAsFactors = TRUE)
```

```
NIWO_Litter_mass <- read.csv(  
  here(processed_data, "NEON_NIWO_Litter_mass_trap_Processed.csv"),  
  stringsAsFactors = TRUE)
```

```
view(NIWO_Litter_mass)  
view(PeterPaul.chem.nutrients)
```

```
#2
```

```
NIWO_Litter_mass <- NIWO_Litter_mass %>%  
  mutate(collectDate = ymd(collectDate))
```

```
PeterPaul.chem.nutrients <- PeterPaul.chem.nutrients %>%  
  mutate(collectDate = ymd(sampledate))
```

```
class(NIWO_Litter_mass$collectDate)
```

```
## [1] "Date"
```

```
class(NIWO_Litter_mass$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
# Define your custom theme
theme_mydefault <- theme_minimal(base_size = 13, base_family = "Helvetica") +
  theme(
    # Customize plot background
    plot.background = element_rect(fill = "gray95", color = NA),

    # Customize plot title
    plot.title = element_text(
      face = "bold",
      size = 16,
      color = "#C8A2C8",
      hjust = 0.5,      # Center title
      margin = margin(b = 10)
    ),

    # Customize axis titles
    axis.title = element_text(face = "bold", color = "#1A5276", size = 12),

    # Customize axis ticks and gridlines
    axis.ticks = element_line(color = "gray40"),
    panel.grid.major = element_line(color = "gray80", size = 0.3),
    panel.grid.minor = element_blank(),

    # Customize legend
    legend.background = element_rect(fill = "white", color = "gray80"),
    legend.title = element_text(face = "bold", color = "#283747"),
    legend.text = element_text(color = "gray")
  )
```

```
## Warning: The 'size' argument of 'element_line()' is deprecated as of ggplot2 3.4.0.
## i Please use the 'linewidth' argument instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
theme_set(theme_mydefault)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
#4

#Total P vs PO4 (fixed aes syntax; remove extra comma)
pp_plot_df <- PeterPaul.chem.nutrients %>%
  filter(lakename %in% c("Peter Lake", "Paul Lake")) %>%
  drop_na(tp_ug, po4)

# checks
nrow(pp_plot_df)

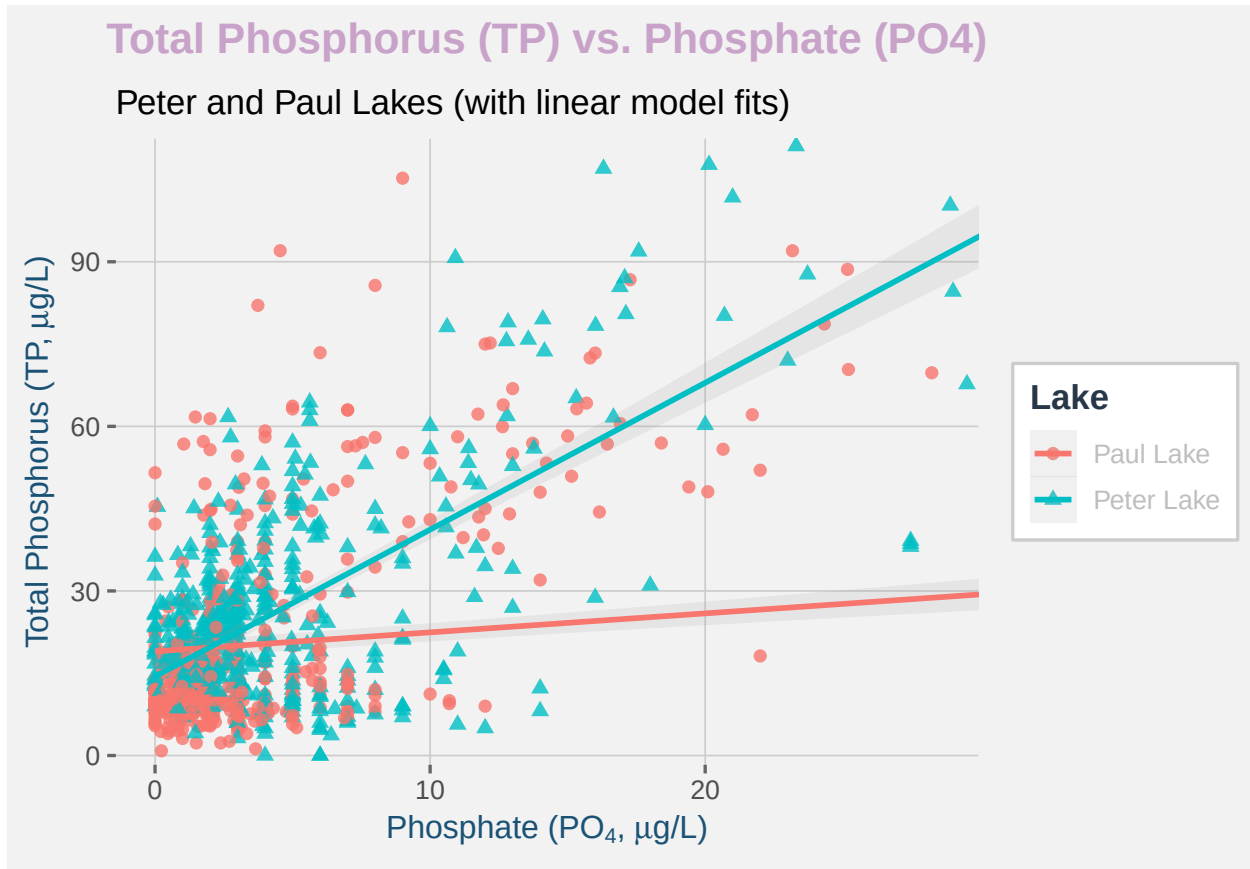
## [1] 1062

table(pp_plot_df$lakename)

##
##   Paul Lake Peter Lake
##      537      525

ggplot(pp_plot_df, aes(x = po4, y = tp_ug, color = lakename, shape = lakename)) +
  geom_point(alpha = 0.8, size = 2) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15) + # no fill mapping -> single legend
  coord_cartesian(
    xlim = quantile(pp_plot_df$po4, c(0.01, 0.99), na.rm = TRUE),
    ylim = quantile(pp_plot_df$tp_ug, c(0.01, 0.99), na.rm = TRUE)
  ) +
  labs(
    title = "Total Phosphorus (TP) vs. Phosphate (PO4)",
    subtitle = "Peter and Paul Lakes (with linear model fits)",
    x = expression(paste("Phosphate (PO"[4], ", ", mu, "g/L)")),
    y = expression(paste("Total Phosphorus (TP, ", mu, "g/L)")),
    color = "Lake",
    shape = "Lake"
  )

## 'geom_smooth()' using formula = 'y ~ x'
```



- [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
PeterPaul.chem.nutrients <- PeterPaul.chem.nutrients %>%
  mutate(
    sampleddate = as.character(sampledate),           # handle factor -> character
    sampleddate = lubridate::parse_date_time(sampledate, orders = c("ymd", "mdy", "dmy")),
    sampleddate = as.Date(sampledate)
  )

pp <- PeterPaul.chem.nutrients %>%
  mutate(
    month = factor(lubridate::month(sampledate, label = TRUE, abbr = TRUE), levels = month.abb),
    lake = stringr::str_trim(as.character(lakename))
  ) %>%
  filter(lake %in% c("Peter Lake", "Paul Lake"))

nrow(pp)
```

```
## [1] 23008
```

```
table(pp$lake)
```

```
##  
## Paul Lake Peter Lake  
##      11060      11948
```

```
small_theme <- theme_minimal(base_size = 12) +  
  theme(  
    plot.background = element_rect(fill = "gray98", color = NA),  
    panel.grid.minor = element_blank(),  
    legend.position = "right",  
    axis.title.x = element_blank()  
  )  
  
# 1. Temperature plot  
p_temp <- ggplot(pp, aes(x = month, y = temperature_C, color = lake)) +  
  geom_boxplot(outlier.size = 1, alpha = 0.6) +  
  labs(y = "Temperature (°C)", title = "(a) Temperature") +  
  scale_color_brewer(palette = "Set1", name = "Lake") +  
  small_theme  
  
# 2. Total Phosphorus plot  
p_tp <- ggplot(pp, aes(x = month, y = tp_ug, color = lake)) +  
  geom_boxplot(outlier.size = 1, alpha = 0.6) +  
  labs(y = expression(paste("TP (", mu, "g/L)")),  
       title = "(b) Total Phosphorus") +  
  scale_color_brewer(palette = "Set1", name = "Lake") +  
  small_theme  
  
# 3. Total Nitrogen plot  
p_tn <- ggplot(pp, aes(x = month, y = tn_ug, color = lake)) +  
  geom_boxplot(outlier.size = 1, alpha = 0.6) +  
  labs(y = expression(paste("TN (", mu, "g/L)")),  
       title = "(c) Total Nitrogen") +  
  scale_color_brewer(palette = "Set1", name = "Lake") +  
  small_theme  
  
legend <- cowplot::get_legend(p_temp + theme(legend.position = "right"))
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;  
## returning the first one. To return all, use 'return_all = TRUE'.
```

```
p_temp_noleg <- p_temp + theme(legend.position = "none")  
p_tp_noleg   <- p_tp   + theme(legend.position = "none")  
p_tn_noleg   <- p_tn   + theme(legend.position = "none")
```

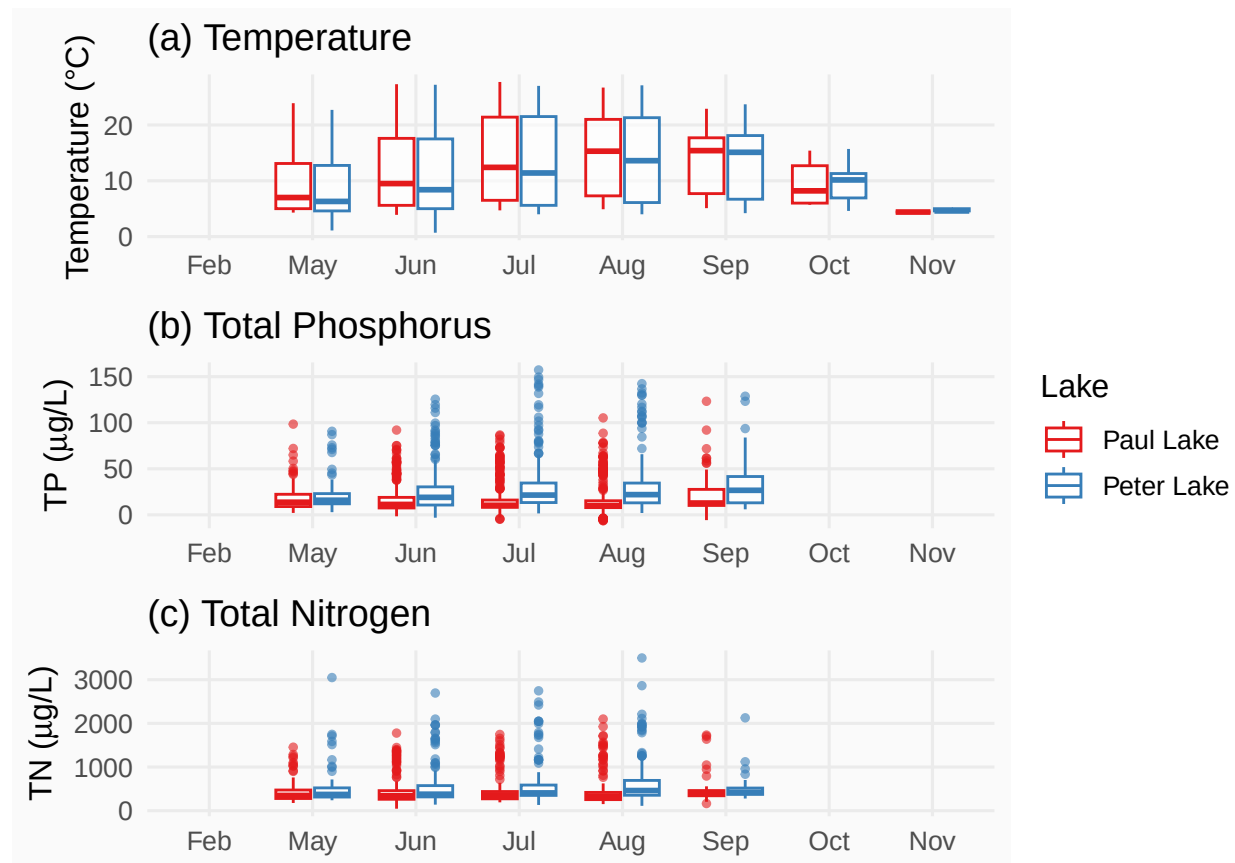
```
plots_col <- cowplot::plot_grid(
  p_temp_noleg, p_tp_noleg, p_tn_noleg,
  ncol = 1, align = "v", axis = "lr", rel_heights = c(1, 1, 1)
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
final_plot <- cowplot::plot_grid(plots_col, legend, ncol = 2, rel_widths = c(1, 0.25))
print(final_plot)
```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: I observed that temperature strongly reflects seasonal changes, while nutrient concentrations (TP and TN) show greater variability between lakes. Peter Lake tends to have higher nutrient levels than Paul Lake, particularly for nitrogen, indicating differing nutrient dynamics or watershed characteristics between the two systems.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
# ensure collectDate is Date (safe conversion)
NIWO_Litter_mass <- NIWO_Litter_mass %>%
  mutate(
    collectDate = as.character(collectDate),
    collectDate = lubridate::parse_date_time(collectDate, orders = c("ymd", "mdy", "dmy")),
    collectDate = as.Date(collectDate)
  )

# inspect functional groups (optional)
unique(NIWO_Litter_mass$functionalGroup)
```

```
## [1] Seeds          Other          Woody material Needles          Leaves
## [6] Twigs/branches Flowers          Mixed
## 8 Levels: Flowers Leaves Mixed Needles Other Seeds ... Woody material
```

```
needles <- NIWO_Litter_mass %>%
  filter(!is.na(collectDate) & !is.na(dryMass)) %>%
  filter(str_to_lower(functionalGroup) == "needles")

# quick check: NLCD classes present
unique(needles$nlcdClass)
```

```
## [1] shrubScrub          grasslandHerbaceous evergreenForest
## Levels: evergreenForest grasslandHerbaceous shrubScrub
```

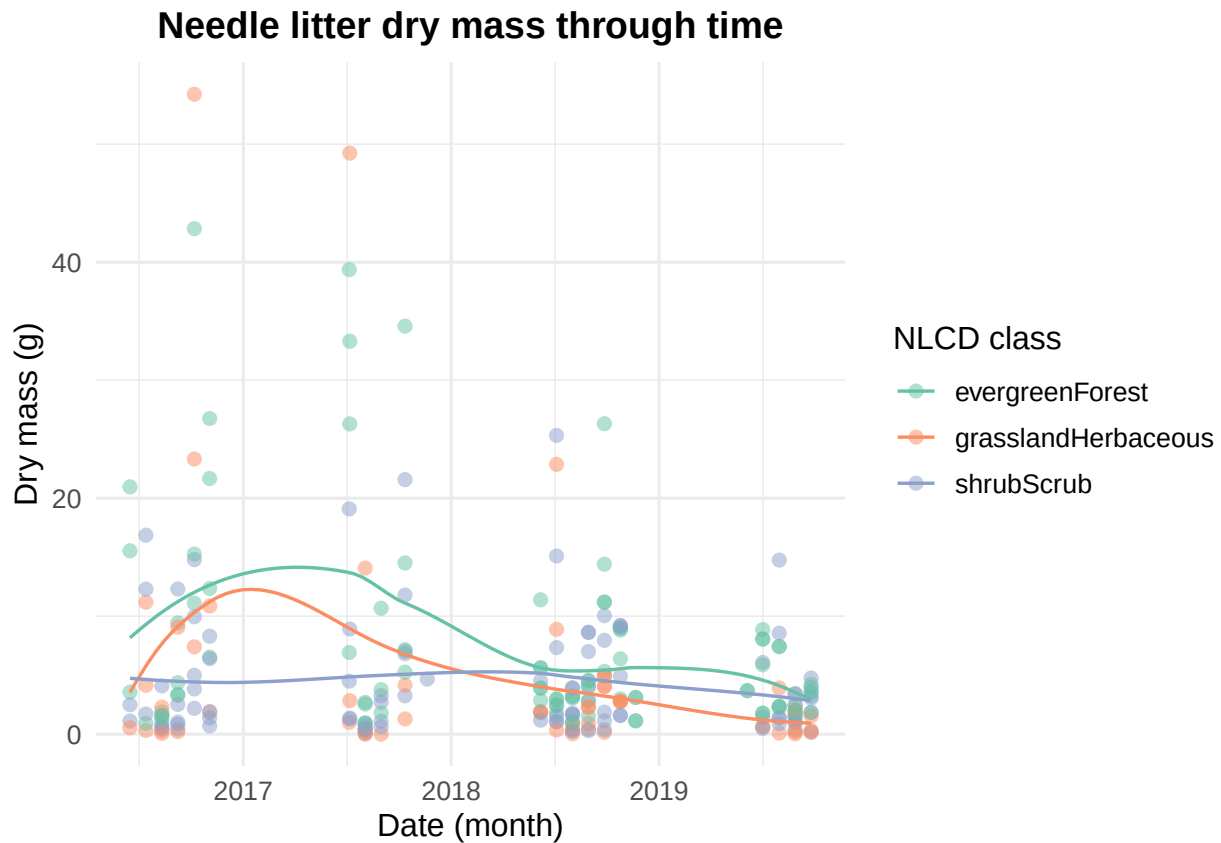
```
p_needles_color <- ggplot(needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.5, size = 2) +
  geom_smooth(aes(group = nlcdClass), method = "loess", se = FALSE, linetype = "solid", size = 0.6) +
  scale_color_brewer(palette = "Set2", name = "NLCD class") +
  labs(
    title = "Needle litter dry mass through time",
    x = "Date (month)",
    y = "Dry mass (g)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "right"
  )
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



```
print(p_needles_color)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
#7
```

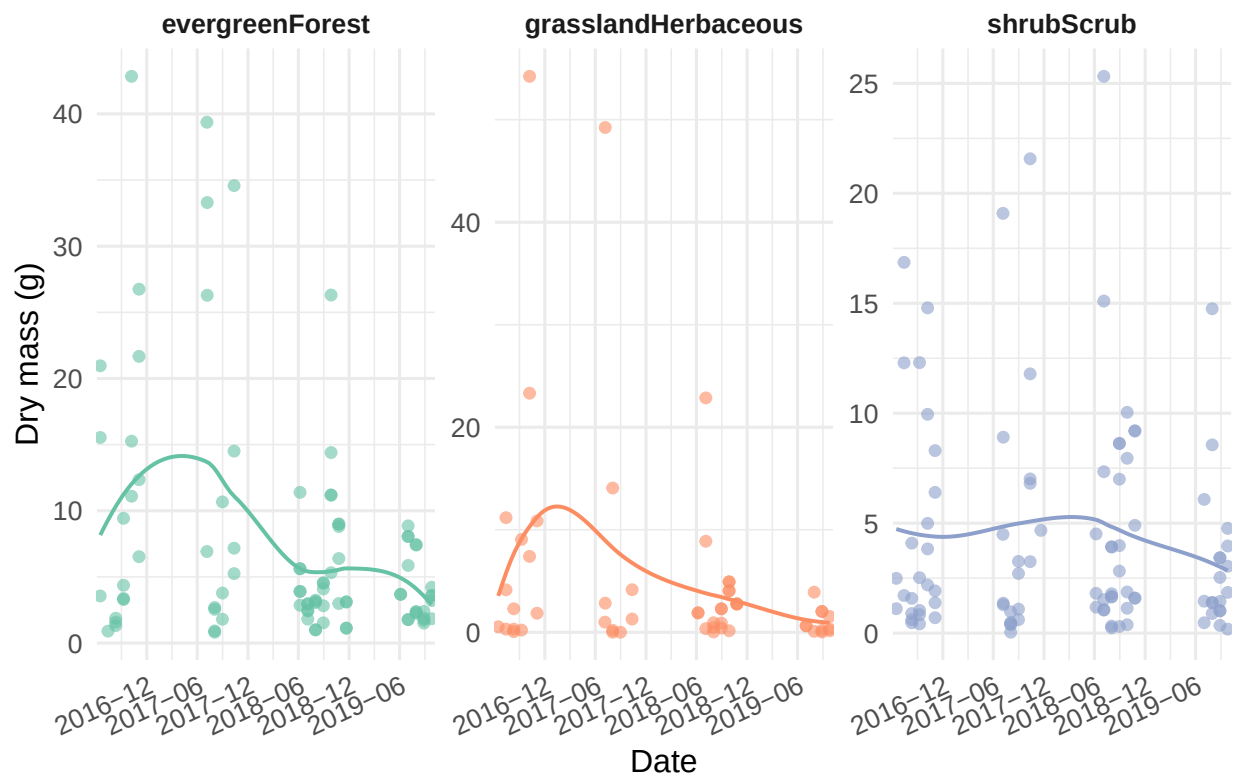
```
p_needles_facets <- ggplot(needles, aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  geom_point(alpha = 0.6, size = 1.6) +
  geom_smooth(method = "loess", se = FALSE, size = 0.7) +
  facet_wrap(~ nlcdClass, ncol = 3, scales = "free_y") +
  scale_color_brewer(palette = "Set2", name = "NLCD class") +
  scale_x_date(date_labels = "%Y-%m", date_breaks = "6 months",
    expand = expansion(mult = c(0.01, 0.01))) +
  labs(
    title = "Needle litter dry mass by NLCD class (faceted, colored by class)",
    x = "Date",
    y = "Dry mass (g)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 25, hjust = 1),
    legend.position = "none" # hide redundant legend if desired
```

```
)

print(p_needles_facets)

## 'geom_smooth()' using formula = 'y ~ x'
```

Needle litter dry mass by NLCD class (faceted, colored by class)



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: Between the two plots, the faceted plot (Plot 7) is more effective in presenting the data. By separating each NLCD class into its own panel, it avoids the overlap of points and trend lines that makes the combined plot (Plot 6) harder to interpret. This layout allows for clearer visualization of the temporal patterns within each land cover type and makes it easier to compare differences in dry mass trends across classes. The consistent axes across panels also support direct visual comparison. In contrast, the combined plot is more cluttered and better suited only for showing overall patterns rather than detailed class-specific trends.